

Tobin A. Driscoll

Professional experience

University of Delaware

- Unidel Chaired Professor of Mathematical Sciences 2023–present
- Professor, Department of Mathematical Sciences 2010–present
- Affiliated Faculty, Department of Biomedical Engineering 2010–present
- Associate Professor, Department of Mathematical Sciences 2004–2010
- Assistant Professor, Department of Mathematical Sciences 2000–2004

University of Colorado at Boulder

- Postdoctoral research fellow, Department of Applied Mathematics 1996–1999

Education

- Ph.D. in Applied Mathematics, Cornell University (supervisor: Lloyd N. Trefethen) 1991–1996
- B.S. in Mathematics with honors, B.S. in Physics, Pennsylvania State University 1987–1991

Honors and awards

- Unidel Chaired Professorship, University of Delaware 2023
- Outstanding Scholar Award, College of Arts & Sciences, University of Delaware 2014
- Winner, 100 Digit Challenge (SIAM) 2002
- NSF VIGRE Postdoctoral Fellow, University of Colorado 1999–2000
- NSF Mathematical Sciences Postdoctoral Research Fellow 1996–1999
- SIAM Outstanding Paper Prize 1999
- Runner-up, Richard C. DiPrima Dissertation Prize 1998
- Second Prize, Leslie Fox Competition 1997
- NSF Graduate Research Fellow 1991–1994

Recent grants

- R. J. Braun (PI), T. A. Driscoll (co-PI). Models for Tear Film Structure, Dynamics and Parameter Identification. NSF DMS-1909846, \$375,000. 2019–2022
- R. J. Braun (PI), T. A. Driscoll, and K. L. Maki (co-PIs). Collaborative Research: Tear Film Dynamics: Modeling, Blinking, and Computation. NSF DMS-1412085, \$375,003. 2014–2017
- M. McCulloch (PI), T. A. Driscoll, and G. Schleiner (co-PIs). Data driven mathematical modeling of the hypoplastic left heart syndrome circulation. NIH Delaware INBRE III, \$60,724. 2014–2016
- T. A. Driscoll. Special support from The MathWorks, Inc. 2013–2015

Books

- T. A. Driscoll and R. J. Braun. *Fundamentals of Numerical Computation*, [online edition](#) in Julia, MATLAB, and Python 2025
- T. A. Driscoll and R. J. Braun. *Fundamentals of Numerical Computation: Julia Edition*, Society for Industrial and Applied Mathematics 2022
- L. N. Trefethen, A. Birkisson, and T. A. Driscoll. *Exploring ODEs*, Society for Industrial and Applied Mathematics 2018

- T. A. Driscoll and R. J. Braun. *Fundamentals of Numerical Computation*, Society for Industrial and Applied Mathematics 2017
- T. A. Driscoll, N. Hale, and L. N. Trefethen, eds. *Chebfun Guide*, 1st edition. Oxford, UK 2014
- T. A. Driscoll, *Learning MATLAB*, Society for Industrial and Applied Mathematics 2009
- T. A. Driscoll and L. N. Trefethen. *Schwarz–Christoffel Mapping*, Cambridge University Press 2002

Book chapters

- R. J. Braun, T. A. Driscoll, and C. G. Begley, “Mathematical Models of the Tear Film,” in *Ocular Fluid Dynamics: Anatomy, Physiology, Imaging Techniques, and Mathematical Modeling*, Springer-Birkhäuser, 2019.
- T. A. Driscoll and B. Fornberg. Padé-based interpretation and correction of the Gibbs phenomenon. In *Advances in the Gibbs Phenomenon*, ed. by A. Jerri, Sigma Sampling Publishing, Potsdam, NY, 2007.
- T. A. Driscoll and L. N. Trefethen. Numerical construction of conformal maps. Appendix to *Fundamentals of Complex Analysis with Applications to Engineering, Science, and Mathematics*, 3rd edition, by E. D. Saff and A. D. Snider, Prentice Hall, 2002.

Refereed articles with over 100 citations (according to Google Scholar as of June 2026)

- Trefethen, L. N., Trefethen, A. E., Reddy, S. C., and Driscoll, T. A. (1993). Hydrodynamic stability without eigenvalues. *Science*, 261(5121), 578–584. [10.1126/science.261.5121.578](https://doi.org/10.1126/science.261.5121.578)
- Driscoll, T. A., and Fornberg, B. (2002). [Interpolation in the limit of increasingly flat radial basis functions](https://doi.org/10.1016/s0898-1221(01)00295-4). *Computers and Mathematics with Applications*, 43(3–5), 413–422. [10.1016/s0898-1221\(01\)00295-4](https://doi.org/10.1016/s0898-1221(01)00295-4)
- Driscoll, T. A. (1996). [Algorithm 756: A MATLAB toolbox for Schwarz-Christoffel mapping](https://doi.org/10.1145/229473.229475). *ACM Transactions on Mathematical Software (TOMS)*, 22(2), 168–186. [10.1145/229473.229475](https://doi.org/10.1145/229473.229475)
- Fornberg, B., Driscoll, T. A., Wright, G., and Charles, R. (2002). [Observations on the behavior of radial basis function approximations near boundaries](https://doi.org/10.1016/s0898-1221(01)00299-1). *Computers and Mathematics with Applications*, 43(3–5), 473–490. [10.1016/s0898-1221\(01\)00299-1](https://doi.org/10.1016/s0898-1221(01)00299-1)
- Driscoll, T. A., and Heryudono, A. R. H. (2007). [Adaptive residual subsampling methods for radial basis function interpolation and collocation problems](https://doi.org/10.1016/j.camwa.2006.06.005). *Computers and Mathematics with Applications*, 53(6), 927–939. [10.1016/j.camwa.2006.06.005](https://doi.org/10.1016/j.camwa.2006.06.005)
- Fornberg, B., and Driscoll, T. A. (1999). [A fast spectral algorithm for nonlinear wave equations with linear dispersion](https://doi.org/10.1006/jcph.1999.6351). *Journal of Computational Physics*, 155(2), 456–467. [10.1006/jcph.1999.6351](https://doi.org/10.1006/jcph.1999.6351)
- Driscoll, T. A., Toh, K.-C., and Trefethen, L. N. (1998). [From Potential Theory to Matrix Iterations in Six Steps](https://doi.org/10.1137/S0036144596305582). *SIAM Review*, 40(3), 547–578. [10.1137/S0036144596305582](https://doi.org/10.1137/S0036144596305582)
- Baggett, J. S., Driscoll, T. A., and Trefethen, L. N. (1995). [A mostly linear model of transition to turbulence](https://doi.org/10.1063/1.868606). *Physics of Fluids*, 7(4), 833–838. [10.1063/1.868606](https://doi.org/10.1063/1.868606)
- Driscoll, T. A., and Fornberg, B. (2001). [A Padé-based algorithm for overcoming the Gibbs phenomenon](https://doi.org/10.1023/A:1016648530648). *Numerical Algorithms*, 26(1), 77–92. [10.1023/A:1016648530648](https://doi.org/10.1023/A:1016648530648)
- Ghrist, M., Fornberg, B., and Driscoll, T. A. (2000). [Staggered time integrators for wave equations](https://doi.org/10.1137/s0036142999351777). *SIAM Journal on Numerical Analysis*, 38(3), 718–741. [10.1137/s0036142999351777](https://doi.org/10.1137/s0036142999351777)
- Driscoll, T. A., and Vavasis, S. A. (1998). [Numerical conformal mapping using cross-ratios and Delaunay triangulation](https://doi.org/10.1137/s1064827596298580). *SIAM Journal on Scientific Computing*, 19(6), 1783–1803. [10.1137/s1064827596298580](https://doi.org/10.1137/s1064827596298580)
- Driscoll, T. A., Bornemann, F., and Trefethen, L. N. (2008). [The chebop system for automatic solution of differential equations](https://doi.org/10.1007/s10543-008-0198-4). *BIT Numerical Mathematics*, 48(4), 701–723. [10.1007/s10543-008-0198-4](https://doi.org/10.1007/s10543-008-0198-4)
- Driscoll, T. A. (1997). [Eigenmodes of isospectral drums](https://doi.org/10.1137/S0036144595285069). *SIAM Review*, 39(1), 1–17. [10.1137/S0036144595285069](https://doi.org/10.1137/S0036144595285069)
- Driscoll, T. A. (2005). [Algorithm 843: Improvements to the Schwarz-Christoffel toolbox for MATLAB](https://doi.org/10.1145/1067967.1067971). *ACM Transactions on Mathematical Software (TOMS)*, 31(2), 239–251. [10.1145/1067967.1067971](https://doi.org/10.1145/1067967.1067971)
- Driscoll, T. A., and Hale, N. (2015). [Rectangular spectral collocation](https://doi.org/10.1093/imanum/dru062). *IMA Journal of Numerical Analysis*, 36(1), 108–132. [10.1093/imanum/dru062](https://doi.org/10.1093/imanum/dru062)
- Platte, R. B., and Driscoll, T. A. (2004). [Computing eigenmodes of elliptic operators using radial basis functions](https://doi.org/10.1016/j.camwa.2003.08.007). *Computers and Mathematics with Applications*, 48(3–4), 561–576. [10.1016/j.camwa.2003.08.007](https://doi.org/10.1016/j.camwa.2003.08.007)
- Platte, R. B., and Driscoll, T. A. (2006). [Eigenvalue stability of radial basis function discretizations for time-dependent problems](https://doi.org/10.1016/j.camwa.2006.04.007). *Computers and Mathematics with Applications*, 51(8), 1251–1268. [10.1016/j.camwa.2006.04.007](https://doi.org/10.1016/j.camwa.2006.04.007)

Refereed articles with 50–100 citations

- Platte, R. B., and Driscoll, T. A. (2005). **Polynomials and Potential Theory for Gaussian Radial Basis Function Interpolation**. SIAM Journal on Numerical Analysis, 43(2), 750–766. [10.1137/040610143](#)
- Heryudono, A., Braun, R. J., Driscoll, T. A., Maki, K. L., Cook, L., and PE, K.-S. (2007). **Single-equation models for the tear film in a blink cycle: Realistic lid motion**. Mathematical Medicine and Biology-a Journal of The Ima, 24(4), 347–377. [10.1093/imammb/dqm004](#)
- Driscoll, T. A. (2010). **Automatic spectral collocation for integral, integro-differential, and integrally reformulated differential equations**. Journal of Computational Physics, 229(17), 5980–5998. [10.1016/j.jcp.2010.04.029](#)
- G. Wojcik, B. Fornberg, R. Waag, J. Mould, T. A. Driscoll, and L. Nikodym. Pseudospectral methods for large-scale bioacoustic models. Proceedings of the 1997 IEEE Ultrasonics Symposium.
- Driscoll, T. A. (2002). **A composite Runge–Kutta method for the spectral solution of semilinear PDEs**. Journal of Computational Physics, 182(2), 357–367. [10.1006/jcph.2002.7127](#)
- Goano, M., Bertazzi, F., Caravelli, P., Ghione, G., and Driscoll, T. A. (2001). **A general conformal-mapping approach to the optimum electrode design of coplanar waveguides with arbitrary cross section**. IEEE Transactions on Microwave Theory and Techniques, 49(9), 1573–1580. [10.1109/22.942569](#)
- Braun, R., Usha, R., McFadden, G., Driscoll, T. A., Cook, L., and King-Smith, P. E. (2012). **Thin film dynamics on a prolate spheroid with application to the cornea**. Journal of Engineering Mathematics, 73(1), 121–138. [10.1007/s10665-011-9482-4](#)
- Heryudono, A. R. H., and Driscoll, T. A. (2010). **Radial basis function interpolation on irregular domain through conformal transplantation**. Journal of Scientific Computing, 44(3), 286–300. [10.1007/s10915-010-9380-3](#)
- Driscoll, T. A., and Fornberg, B. (1999). **Block pseudospectral methods for Maxwell’s equations II: Two-dimensional, discontinuous-coefficient case**. SIAM Journal on Scientific Computing, 21(3), 1146–1167. [10.1137/s106482759833320x](#)
- Pelesko, J. A., and Driscoll, T. A. (2006). **The effect of the small-aspect-ratio approximation on canonical electrostatic MEMS models**. Journal of Engineering Mathematics, 53(3–4), 239–252. [10.1007/s10665-005-9013-2](#)
- Birkisson, A., and Driscoll, T. A. (2012). **Automatic Fréchet differentiation for the numerical solution of boundary-value problems**. ACM Transactions on Mathematical Software, 38(4), 1–29. [10.1145/2331130.2331134](#)

Refereed articles with 20–49 citations

- L. N. Trefethen and T. A. Driscoll. Schwarz–Christoffel mapping in the computer era. Proceedings of the International Congress of Mathematicians, Vol. III (Berlin, 1998). *Doc. Math.* 1998, Extra Vol. III, 533–542 (electronic).
- Driscoll, T. A., and Trefethen, L. N. (1996). **Pseudospectra for the wave equation with an absorbing boundary**. Journal of Computational and Applied Mathematics, 69(1), 125–142. [10.1016/0377-0427\(95\)00021-6](#)
- DeLillo, T. K., Driscoll, T. A., Elcrat, A. R., and Pfaltzgraff, J. A. (2008). **Radial and circular slit maps of unbounded multiply connected circle domains**. Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, 464(2095), 1719–1737. [10.1098/rspa.2008.0006](#)
- Driscoll, T. A., Nakatsukasa, Y., and Trefethen, L. N. (2024). **AAA rational approximation on a continuum**. SIAM Journal on Scientific Computing, 46(2), A929–A952, 2024. [10.1137/23M1570508](#)
- Driscoll, T. A., and Fornberg, B. (1998). **A Block Pseudospectral Method for Maxwell’s Equations: I. One-Dimensional Case**. J. Comput. Phys., 140, 1–19. [10.1006/jcph.1998.5883](#)
- Braun, R. J., Driscoll, T. A., Begley, C. G., King-Smith, P. E., and Siddique, J. I. (2018). **On tear film breakup (TBU): Dynamics and imaging**. Mathematical Medicine and Biology, 35(2), 145–180. [10.1093/imammb/dqw023](#)
- Maki, K. L., Braun, R. J., Driscoll, T. A., and King-Smith, P. E. (2008). **An overset grid method for the study of reflex tearing**. Math. Med. Biol., 25, 187–214. [10.1093/imammb/dqn013](#)
- Driscoll, T. A., and Maki, K. L. (2007). **Searching for rare growth factors using multicanonical Monte Carlo methods**. SIAM Review, 49(4), 673–692. [10.1137/050637662](#)
- Li, L., Braun, R. J., Driscoll, T. A., Henshaw, W. D., Banks, J. W., and King-Smith, P. E. (2016). **Computed tear film and osmolarity dynamics on an eye-shaped domain**. Mathematical Medicine and Biology, 33(2), 123–157. [10.1093/imammb/dqv013](#)

- Deng, Q., Braun, R. J., Driscoll, T. A., and King-Smith, P. E. (2013). [A model for the tear film and ocular surface temperature for partial blinks](#). *Interfacial Phenomena and Heat Transfer*, 1(4), 357–381. [10.1615/interfacphenomheattransfer.v1.i4.40](#)
- Usher, D. C., Driscoll, T. A., Dhurjati, P., Pelesko, J. A., Rossi, L. F., Schleininiger, G., Pusecker, K., and White, H. B. (2010). [A transformative model for undergraduate quantitative biology education](#). *CBE Life Sciences Education*, 9(3), 181–188. [10.1187/cbe.10-03-0029](#)
- DeLillo, T. K., Driscoll, T. A., Elcrat, A. R., and Pfaltzgraff, J. A. (2006). [Computation of multiply connected Schwarz-Christoffel maps for exterior domains](#). *Computational Methods and Function Theory*, 6(2), 301–315. [10.1007/BF03321616](#)
- Deng, Q., Braun, R., and Driscoll, T. A. (2014). [Heat transfer and tear film dynamics over multiple blink cycles](#). *Physics of Fluids*, 26(7), 071901. [10.1063/1.4887341](#)

Recent refereed proceedings

- Tichenor, A. A., P. Situ, T. A. Driscoll, C. G. Begley, and R. J. Braun. “Linking TBU Mechanisms and the Neurosensory Response in Dry Eye Subjects.” *Investigative Ophthalmology & Visual Science* 65, no. 7 (2024): 6572–6572.
- Situ, P., A. A. Tichenor, T. A. Driscoll, C. G. Begley, and R. J. Braun. “Neurosensory Function Changes Induced by Tear Film Instability (TFI)-Related Stress in Dry Eye: A Pilot Study.” *Investigative Ophthalmology & Visual Science* 65, no. 7 (2024): 6566–6566.
- Driscoll, T. A., A. A. Tichenor, P. Situ, R. J. Braun, and C. G. Begley. “Optimized Models of Tear Breakup (TBU) Mechanisms in Dry-Eye (DE) and Non-DE Subjects.” *Investigative Ophthalmology & Visual Science* 65, no. 7 (2024): 6559–6559.
- R. J. Braun, T. Driscoll, C. Begley, P. Situ, A. Tichenor, and R. Luke, “Tear Breakup (TBU) Analysis with Fluorescence (FL) and Thermal (TH) imaging,” *Investigative Ophthalmology & Visual Science*, vol. 64, no. 8, pp. 186–186, Jun. 2023.
- R. J. Braun, T. A. Driscoll et al., “Data and Analysis from Tear Breakup (TBU) in Normal Subjects,” *Investigative Ophthalmology & Visual Science*, vol. 63, no. 7, pp. 3950-A0230, Jun. 2022.
- R. A. Luke et al., “Fitting Simplified Models to Machine Learning-Identified Tear Film Breakup,” in *Investigative Ophthalmology & Visual Science*, Jun. 2021, vol. 62, p. 1315.

Software

- T. A. Driscoll. [RationalFunctionApproximation.jl](#). Approximation by rational functions in Julia. 2025–
- T. A. Driscoll. [ComplexRegions.jl](#). Complex regions and paths in Julia. 2019–
- L. N. Trefethen and others, [Chebfun](#). Numerical computing with functions. 2008–2014
- T. A. Driscoll. [Schwarz–Christoffel Toolbox for MATLAB](#). Conformal maps to regions bounded by polygons. 1994–2002

Graduate students supervised

Yuxing Zhou, Ph. D. candidate; Arnab Roy, Ph. D. candidate; Qinying Chen, Ph. D. expected 2026; Kevin Aiton, Ph.D.; Shawn Abernethy, M.S. with thesis; Quan Deng, Ph.D.; Alfa Heryudono, Ph.D.; Rodrigo Platte, Ph.D.

Service highlights

Delaware

- Director of Graduate Studies (Delaware), 2011–2014
- Director of Undergraduate Studies (Delaware), 2018–2021
- Chaired ad hoc committee to create B.S. degree in Data Science
- Director of the Master of Science program in Data Science, 2023–2025
- ADVANCE Fellow and Senior Fellow, 2020–2023
- University Strategic Planning Working Group on Redefining Creativity, Innovation and Entrepreneurship, 2021

Profession

- Editor-in-Chief of the SIAM book series on Software, Environments, and Tools, 2026–present
- Communicating Editor of *Advances in Computational Mathematics*, 2019–present

- At-Large Member of the SIAM Council, 2015–2018
- Associate Editor of *Journal of Engineering Mathematics*, 2010–2015
- Associate Editor of the *SIAM Journal on Scientific Computing*, 2008–2014